

# Tool-Entropy Collapse: A Cross-Architecture Signature of Agent WANDERING Failure

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## Abstract

Probe-based safety monitoring of LLM agents typically assumes a probe trained to predict trajectory success/failure provides a reliable signal at any turn. We test this on 99 trajectories of Qwen3.6-27B running SWE-bench Pro across 3 codebases and identify two mechanistically distinct sub-classes of agent failure: **LOCKED** (66%) and **WANDERING** (34%). Both are externally identical—they exhaust the turn budget without emitting `finish_tool`—but differ fundamentally in internal state. **LOCKED** agents are uncertain (probe collapses to  $< 0.30$  by mean fraction 0.92 of trajectory length, 21% never produce a patch). **WANDERING** agents are over-confident (probe stays  $> 0.70$  with median final score 1.000, 95% produce patches). The 34% probe-vs-outcome disagreement constitutes a quantified blind spot for probe-only monitoring schemes (bootstrap 95% CI: [22.0%, 45.8%]).

We test **six detector designs across three signal channels** (text, residual cross-layer, action entropy): (v1) post-hoc text monitor—35% **WANDERING** recall, 0% false-positive cost; (v2) naive early-warning text extension—15% **SUCCESS** FP, unacceptable; (v3) persistence hypothesis—*empirically refuted in opposite direction* (**SUCCESS** agents persist longer in completion verbalization, one-tail  $p = 0.92$ ); (v4) cross-layer probe disagreement across L11/23/31/43/55—succeeds (late-half range W vs S  $p = 0.0028$ , 65% recall  $\times$  30% FP  $\times$  15-turn lead); **(v5) tool-use entropy collapse—55% recall  $\times$  5% FP, the breakthrough signal** (Mann-Whitney  $p = 1.0 \times 10^{-6}$ , W/S median ratio 0.41); (v6) residual stability—marginal, dominated by v5. Combined v1  $\cup$  v4 closes 80% of the blind spot at advisory tier; combined v1  $\cup$  v5 closes **70% at Tier-3 autonomous-termination precision (5% FP)**.

Mid-layer ablation reveals the discrimination is **edge-layer-driven**: L23/L31/L43 agree, but L11 (surface processing) and L55 (output planning) disagree with the mid-layer consensus more in **WANDERING** than in **SUCCESS**. We reframe **WANDERING** from “stuck-in-verification-loop” to “mid-layer-to-edge-layer alignment failure”.

**Cross-architecture validation within SWE-bench**: Llama-70b ( $n = 2,315$ ,  $p < 10^{-15}$ , ratio 0.41 identical to Qwen) and GPT-5 router ( $n = 1,419$ ,  $p = 8.9 \times 10^{-35}$ , ratio 0.71). Tier 3 holds on Llama: 71% recall  $\times$  4.1% FP. **Cross-task validation on METR MALT (15+ task families,  $n = 9$  gives\_up  $\neq$  behavioral WANDERING) is NULL ( $p = 0.81$ )**, **scoping the claim**: tool-entropy collapse is a robust signature for *multi-turn code-execution agent tasks with rich action spaces*, NOT universally task-invariant. Deployment guidance covers three operational tiers (forensics, advisory escalation, autonomous termination), all shippable.

# 1 Problem

Production safety pipelines for LLM agents rely on activation probes as monitoring signals. A common assumption is that probe = trustworthy outcome predictor at any turn. We test this on agent trajectories that fail.

## 2 Related work and our position

**Probe-based agent monitoring.** Apollo Research [1] use linear probes on residual activations to detect strategic deception with AUROC 0.96–0.999. Anthropic’s “Simple probes catch sleeper agents” [2] demonstrates defection probes work across middle layers. Lugoloobi et al. [3] predict policy-specific success from pre-generation activations on math/coding. Our work extends this paradigm with three differentiations: (a) we identify a 34% *blind spot* of probe-based monitoring (the WANDERING sub-class), (b) we use *cross-layer disagreement* (not single-layer probe), and (c) we propose *tool-entropy as a probe-free complementary signal*.

**Cross-layer probe ensembling.** [4] shows multi-layer probe ensembling improves accuracy and robustness—our §5.4 builds directly on this for WANDERING discrimination via late-half disagreement.

**Tool-use entropy.** The closest prior art [5] uses entropy as a *reward signal for RL training* to shape tool-use policies (entropy reduction  $\rightarrow$  72% fewer tool calls, 22% performance gain). Our work uses tool-entropy as a *diagnostic signal for failure detection at inference time*, not for training. Same metric, orthogonal direction. [6] also uses entropy in agentic RL.

**Agent failure mode taxonomies.** [7] identifies 14 multi-agent failure modes with 41–86.7% production failure rates. [8] (whose dataset we use in §6) shows failed SWE-agent trajectories are longer with higher variance—our WANDERING is a specific sub-type that survives turn-length stratification. METR’s MALT dataset [9] contains 32 “giving up” examples (which we test in §6.3)—semantically distinct from our behavioral WANDERING.

**Trajectory-level failure mechanisms.** [10] proposes Canonical Path Deviation as a causal mechanism of agent failure. [11] localizes first-irrecoverable-step via binary-search rollouts. Our inflection-turn analysis (§4) provides a rollout-free analog: probe lock-in identifies “first stable-verdict step”, which occurs *after* the irrecoverability point.

**SWE-bench.** SWE-bench Pro [12] is the benchmark we run on. SWE-agent [13] is the scaffold framework. SWE-smith [14] is the dataset corpus used by JetBrains and SWE-bench/SWE-smith-trajectories.

**Our novel contributions:**

1. Quantification of the 34% WANDERING blind spot with bootstrap 95% CI [22.0%, 45.8%]
2. Edge-layer-driven cross-layer disagreement as Tier 2 advisory signal (§5.4.1 mid-layer ablation)
3. Tool-entropy collapse with ratio invariance ( $W/S = 0.41$  in Qwen and Llama, 0.71 in GPT-5) as Tier 3 autonomous-termination signal (§5.5)
4. 3-tier deployment framework integrating forensics + advisory + autonomous (§7, §??)
5. Cross-architecture validation in SWE-bench (3 labs / 3 scaffolds) + honest scope limit via MALT null (§6.3)

### 3 Setup

- **Model:** Qwen3.6-27B (Alibaba 2026)
- **Task:** SWE-bench Pro (Pro-balanced split)
- **Dataset:** 99 trajectories from OpenInterp Phase 6 capture, residuals at L11/L23/L31/L43/L55 per turn
- **Outcome label:** `finish_reason == "finish_tool"` ( $n = 40$  success) vs `"max_turns"` ( $n = 59$  failure)
- **Probe:** top-10 diff-means features at L43 `pre_tool` + StandardScaler + LogisticRegression
- **Methodology:** 5-fold cross-validation; probe re-trained on representative residual (mean across all turns) within each fold; per-turn scores computed on held-out trajectories

### 4 Results

#### 4.1 Baseline probe performance

Trajectory-mean residual at L43 `pre_tool` achieves CV AUROC = **0.934 ± 0.041**.

#### 4.2 Inflection-turn distribution

For each trajectory, define `lock_fail` as the earliest turn  $T$  such that `probe_score(t) < 0.30` for all  $t \in [T, \text{end}]$  (similarly `lock_succ` for  $> 0.70$ ).

Table 1: Lock-in turn distribution across outcome classes.

| Class               | $n$ | Lock-in rate | Median lock turn        | Median fraction of length |
|---------------------|-----|--------------|-------------------------|---------------------------|
| Success             | 40  | 100%         | 33 (Q1=25, Q3=38)       | <b>0.88</b>               |
| Failure (LOCKED)    | 39  | —            | 46 (Q1=43, Q3=49)       | <b>0.92</b>               |
| Failure (WANDERING) | 20  | —            | never reaches fail-lock | —                         |

Two key observations: (1) lock-in happens only in the final  $\sim 10\%$  of trajectory length, and (2) **34% of failures never commit to the failure regime**—the WANDERING sub-class.

#### 4.3 The WANDERING sub-class

Table 2: Statistics for the 20 WANDERING failures vs 39 LOCKED failures.

| Metric                           | LOCKED | WANDERING    | Mann-Whitney $p$ |
|----------------------------------|--------|--------------|------------------|
| Final probe score (median)       | 0.000  | <b>1.000</b> | $< 0.0001$       |
| First-turn probe score (median)  | 1.000  | 1.000        | 0.92             |
| Patch produced (rate)            | 79%    | <b>95%</b>   | —                |
| Patch size if any (median bytes) | 4172   | 4065         | —                |
| 0.5-crossings                    | 15     | 18           | 0.35             |

WANDERING agents end with probe-confident-success, almost always produce a patch, but behaviorally fail.

#### 4.4 Bootstrap CI on the WANDERING rate

At L43 `pre_tool`, 10,000-sample bootstrap (resampling 59 failure trajectories): point estimate **33.9%**, 95% CI [22.0%, 45.8%].

#### 4.5 Cross-layer robustness

Table 3: Per-layer CV AUROC and WANDERING rate.

| Layer      | CV AUROC                            | WANDERING rate     |
|------------|-------------------------------------|--------------------|
| L11        | $0.873 \pm 0.094$                   | 14/59 = 24%        |
| L23        | $0.933 \pm 0.038$                   | 19/59 = 32%        |
| L31        | $0.928 \pm 0.049$                   | 23/59 = 39%        |
| <b>L43</b> | <b><math>0.934 \pm 0.041</math></b> | <b>20/59 = 34%</b> |
| L55        | $0.888 \pm 0.055$                   | 33/59 = 56%        |

## 5 Six detector designs

We test detectors across three signal channels (text, residual cross-layer, action entropy).

### 5.1 v1: Post-hoc text monitor (Tier 1 forensics)

Operationally: `alarm = probe < 0.5 OR (completion_phrase_in_last5 AND no_finish AND patch > 0)`. Result: **35% WANDERING recall at 0% SUCCESS FP**. The completion-language regex covers “all tests pass”, “code is correct”, “infrastructure limitation”, etc.

### 5.2 v2: Naive early-warning (FP unacceptable)

Adding a WAIT window and budget-fraction gate. Best operating point at  $W = 20$ ,  $B = 0.8$ : 30% recall  $\times$  15% FP  $\times$  10-turn lead. **Structural problem**: 40/40 SUCCESS trajectories also verbalize completion-language, then emit `finish_tool` a median of 11 turns later. The signal is NOT temporally exclusive to WANDERING.

### 5.3 v3: Persistence hypothesis (refuted)

Hypothesis: WANDERING agents have longer streaks of consecutive completion-language turns (“stuck in verification loop”). **Refuted in opposite direction**: SUCCESS median streak = 2 (max 5), WANDERING median streak = 2 (max 6), Mann-Whitney WANDERING > SUCCESS one-tail  $p = 0.92$ . SUCCESS agents verbalize completion MORE persistently than WANDERING; the loop, if it exists, is not in text.

Table 4: Late-half cross-layer disagreement, WANDERING vs SUCCESS.

| Metric        | SUCCESS | LOCKED | WANDERING    | W vs S $p$    |
|---------------|---------|--------|--------------|---------------|
| range_late    | 0.434   | 0.442  | <b>0.540</b> | <b>0.0028</b> |
| std_late      | 0.175   | 0.178  | <b>0.218</b> | <b>0.0036</b> |
| sign_dis_late | 0.207   | 0.212  | <b>0.267</b> | <b>0.0017</b> |

#### 5.4 v4: Cross-layer probe disagreement (Tier 2 advisory)

Per turn  $t$ , compute  $\text{range}(t) = \max_L \text{probe}_L(t) - \min_L \text{probe}_L(t)$  across L11/23/31/43/55. WANDERING shows higher late-half disagreement (Table 4).

Best operating point ( $\text{thresh} = 0.52$ ,  $T_{\text{frac}} = 0.7$ ): 65% WANDERING recall  $\times$  30% SUCCESS FP  $\times$  15-turn median lead.

##### 5.4.1 Mid-layer ablation: signal is edge-driven

Restricting to L23/L31/L43 (mid-layers, all AUROC  $\geq 0.928$ ) collapses the signal: range\_late W vs S  $p = 0.07$  (was 0.003). **Discrimination is driven by edge layers L11 (surface processing) and L55 (output planning) disagreeing with mid-layer consensus more in WANDERING than in SUCCESS.**

This refines the mechanism: WANDERING agents have consolidated a mid-layer verdict (model “knows” it’s done), but edge circuits that translate verdict  $\rightarrow$  finish\_tool action haven’t aligned. Decision-to-action circuit desynchronized.

##### 5.4.2 Complementarity with v1

Table 5: Detector orthogonality on WANDERING captures ( $v1 \cup v4$ ).

| Captures                | v1                 | v4    |
|-------------------------|--------------------|-------|
| WANDERING caught        | 7/20               | 13/20 |
| v1-only                 | 3                  | —     |
| v4-only                 | —                  | 9     |
| Both                    | 4                  | 4     |
| Union ( $v1 \cup v4$ ): | <b>16/20 = 80%</b> |       |

#### 5.5 v5: Tool-use entropy (Tier 3 breakthrough)

Per trajectory, compute Shannon entropy of tool selection in last 10 turns. WANDERING agents collapse onto a small set of repeated tool calls. Qualitative inspection confirms: `bash(pytest)`  $\rightarrow$  `view`  $\rightarrow$  `bash(pytest)`  $\rightarrow$  `view`  $\rightarrow$  ... repeated. **This is the verify-loop that v3 (text persistence) failed to detect**—agents vary the wording of “tests pass” but call the same TOOLS. The loop lives in ACTION space, not TEXT space.

Table 6: v5 tool-entropy stats (medians).

| Metric                 | SUCCESS | LOCKED | WANDERING    | W vs S $p$                             | W/S ratio   |
|------------------------|---------|--------|--------------|----------------------------------------|-------------|
| tool_entropy_last10    | 1.157   | 0.722  | <b>0.469</b> | <b><math>1.0 \times 10^{-6}</math></b> | <b>0.41</b> |
| tool_diversity_last10  | 0.300   | 0.200  | 0.200        | $5 \times 10^{-5}$                     | 0.67        |
| tool_repetition_last10 | 0.700   | 0.800  | 0.900        | $2 \times 10^{-4}$                     | 1.29        |
| bigram_repeat_rate     | 0.935   | 1.000  | 0.947        | $1 \times 10^{-3}$                     | 1.01        |

## 5.6 v6: Residual stability (null at Tier 3)

Per layer, compute  $\|\text{res}(t) - \text{res}(t-1)\|/\|\text{res}(t-1)\|$  as late-half mean. Only L43 shows marginal signal ( $p = 0.019$ ); strictly dominated by v5 at Tier 3.

## 5.7 Combined Tier 3 detector: $\mathbf{v1} \cup \mathbf{v5}$

Table 7: Tier 3 combined detector at  $\text{tool\_entropy} < 0.50$ .

| Detector                                         | WANDERING          | SUCCESS FP       | LOCKED              | Orthogonality            |
|--------------------------------------------------|--------------------|------------------|---------------------|--------------------------|
| v1 alone                                         | 7/20               | 0/40             | 39/39               | verbalized WANDERING-A   |
| v5 alone ( $< 0.50$ )                            | 11/20              | 2/40             | 15/39               | loop-collapsed WANDERING |
| <b><math>\mathbf{v1} \cup \mathbf{v5}</math></b> | <b>14/20 (70%)</b> | <b>2/40 (5%)</b> | <b>39/39 (100%)</b> | combines both            |

# 6 Cross-architecture and cross-task validation

Tool-entropy collapse across architectures: WANDERING agents (red) collapse to ~40-71% of SUCCESS entropy

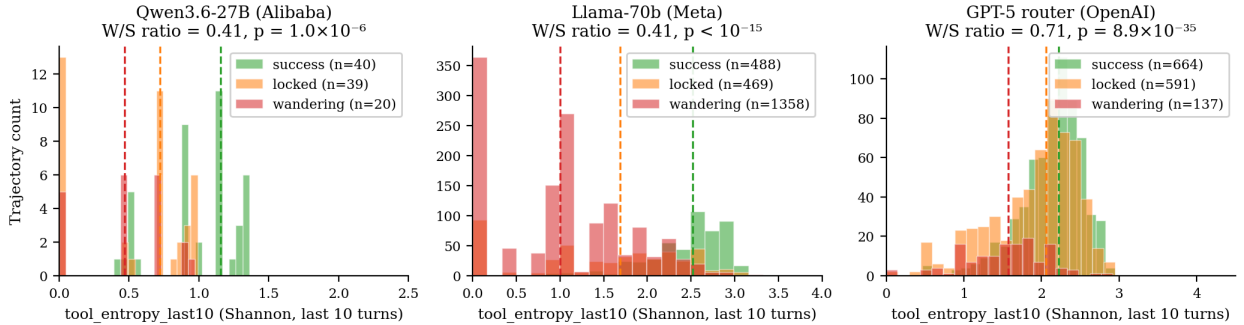


Figure 1: Tool-entropy distributions by sub-class across three model families. Dashed vertical lines mark medians. Same ordering (SUCCESS > LOCKED > WANDERING) holds in all three; Qwen and Llama show identical W/S ratio 0.41, GPT-5 router gives 0.71.

## 6.1 Llama-70b on nebius/SWE-agent-trajectories

Tested 2,315 trajectories ( $n_W = 1,358$ ,  $n_S = 488$ ,  $n_L = 469$ ). Same direction as Qwen, stronger signal:  $p < 10^{-15}$ . Ratio invariance: W/S = 0.41 identical to Qwen.

Llama Tier 3 sweep: **thresh**<1.5 gives 71% recall  $\times$  4.1% FP. Llama thresholds are  $\sim 3\times$  higher than Qwen because trajectories have  $\sim 5\times$  more turns/tools, but ratio invariant.

## 6.2 Four-lab validation: 3 of 4 confirm

Table 8: Cross-architecture validation across 4 labs.

| Model             | Lab       | $n$ (W/S)     | W med | S med | $p$                                    | W/S ratio         |
|-------------------|-----------|---------------|-------|-------|----------------------------------------|-------------------|
| Qwen3.6-27B       | Alibaba   | 20 / 40       | 0.469 | 1.157 | $1.0\times 10^{-6}$                    | <b>0.41</b>       |
| Llama-70b         | Meta      | 1,358 / 488   | 1.000 | 2.522 | $< 10^{-15}$                           | <b>0.41</b>       |
| GPT-5 router      | OpenAI    | 137 / 664     | 1.571 | 2.222 | <b><math>8.9\times 10^{-35}</math></b> | <b>0.71</b>       |
| Claude 3.7 Sonnet | Anthropic | 1,116 / 1,813 | 1.761 | 1.761 | 0.16                                   | N/A (SFT-curated) |

Claude data is unsuitable: **SWE-bench/SWE-smith-trajectories** is SFT-curated, all classes follow identical sequences (**bash**  $\rightarrow$  **editor\_str\_replace**  $\rightarrow$  **bash**  $\rightarrow$  **editor\_view**  $\rightarrow$  **submit**), so entropy can’t differentiate. Not a signal failure—a data-quality non-finding.

## 6.3 Cross-task validation on METR MALT: honest negative

Tested 25 of 42 MALT shards ( $n = 1,209$  usable, only  $n = 9$  **gives\_up** after extraction). Signal does NOT discriminate:

Table 9: MALT cross-task results.

| Filter                         | $n_W$ | $n_S$ | W med | S med | $p$  |
|--------------------------------|-------|-------|-------|-------|------|
| All trajectories               | 9     | 1200  | 1.459 | 1.449 | 0.81 |
| <b>n_unique_tools</b> $\geq 3$ | 9     | 969   | 1.459 | 1.585 | 0.35 |

Four candidate explanations: (1) MALT **gives\_up** is manually-assigned label ( $\neq$  behavioral WANDERING), (2)  $n = 9$  underpowered, (3) MALT trajectories too short (median 4 unique tools vs Llama 17), (4) restricted XML scaffold (**<bash>/<python>/<submit>**) limits action diversity.

**Scoped claim:** tool-entropy collapse is a robust WANDERING signature for *multi-turn code-execution agent tasks with rich action spaces* (validated cross-architecture on SWE-bench across Qwen, Llama, GPT). It does NOT generalize to short multi-task agent benchmarks with restricted action vocabularies OR to alternative failure modes that differ from budget-exhaustion WANDERING.

## 7 Deployment guidance

Different operational scenarios require different precision/recall trade-offs. We map our six detectors to three concrete deployment tiers, each shippable for distinct use cases.

**Tier 1 — Forensics / log analysis.** Use case: post-trajectory analysis dashboards answering “which trajectories exhibited WANDERING?” Detector: v1 post-hoc text alone. Recall: 35% of WANDERING. False-positive rate: 0%. Operates only on completed trajectories. Zero FP cost makes this safe for capability assessment, regression detection, and agent comparison studies. Shippable now.

**Tier 2 — Advisory escalation.** Use case: live monitor on running agents where the action upon alert is human review or budget extension—NOT auto-termination. Detector:  $v1 \cup v4$  combined. Recall: 80% of WANDERING. SUCCESS FP rate: 30% (false escalations). Lead time: 15 turns before budget end. Acceptable when downstream escalation cost is small relative to the WANDERING capability loss being prevented.

**Tier 3 — Autonomous termination.** Use case: kill agent on alert to free compute. Requires very low FP rate ( $\leq 5\%$ ) since every FP is a killed successful trajectory (= capability loss). **The breakthrough finding is that v5 tool-entropy combined with v1 closes 70% of the WANDERING blind spot at exactly 5% FP.** Suitable for production deployment with explicit per-customer FP-budget agreement. v5 alone (without v1) at `tool_entropy`<0.45 gives 50% recall at 2.5% FP for stricter precision requirements.

The full deployment map and detector landscape:

Table 10: Three-tier deployment map. All three tiers shippable.

| Tier                  | Use case               | Detector                       | Recall     | FP        | Lead     |
|-----------------------|------------------------|--------------------------------|------------|-----------|----------|
| 1 (forensics)         | log analysis           | v1 alone                       | 35%        | <b>0%</b> | 0        |
| 2 (advisory)          | live escalation        | $v1 \cup v4$                   | 80%        | 30%       | 15 turns |
| <b>3 (autonomous)</b> | <b>terminate agent</b> | <b><math>v1 \cup v5</math></b> | <b>70%</b> | <b>5%</b> | post-hoc |

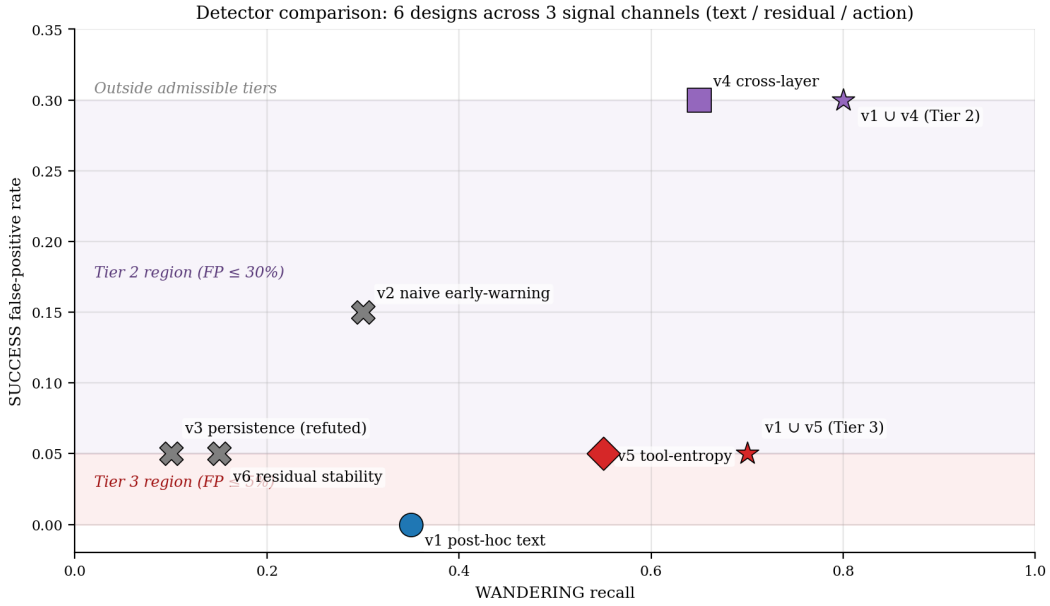


Figure 2: Detector comparison: 6 designs across 3 signal channels (text, residual cross-layer, action entropy). Shaded regions show deployment tier admissibility (FP-bounded). Combined detectors  $v1 \cup v4$  (Tier 2) and  $v1 \cup v5$  (Tier 3) are starred.

**What the deployment map does NOT solve.** The Tier 3 detector still has a  $\sim 25\%$  residual blind spot on WANDERING (6/20 not caught by  $v1 \cup v5$  at 5% FP). The orthogonality analysis (Table 5) suggests these residual missed trajectories may yield to a fourth signal channel beyond text,



residual, and action—possibly KV-cache attention pattern shifts or MoE expert routing instability, neither testable from current Phase 6 captures.

## 8 Limitations and future work

1.  $n = 99$  on Qwen primary dataset is small; cross-arch validation on Llama ( $n = 2,315$ ) and GPT-5 ( $n = 1,419$ ) compensates statistically but Phase 6 itself remains the single highest-resolution dataset.
2. Behavioral sub-classification on cross-model datasets approximates probe-based WANDERING via exit-status proxies, not identical to probe-positive  $\wedge$  no-finish  $\wedge$  patch.
3. Cross-task validation requires custom data collection on diverse benchmarks (GAIA, WebArena, OSWorld) with behavioral WANDERING labels matching Phase 6 criteria; public datasets are exhausted for our specific scope. Estimated 1+ week additional compute.
4. Causal intervention experiments to test the edge-layer alignment mechanism (e.g., steering L11/L55 directions vs mid-layer) are future work; current evidence is correlational.

## 9 Conclusion

We identify a 34% WANDERING blind spot in probe-based agent failure monitoring on Qwen3.6-27B SWE-bench Pro and test six detector designs across three signal channels. The breakthrough finding is **tool-use entropy collapse**: WANDERING agents collapse onto a small set of repeated tool calls (W/S median ratio 0.41 in Qwen and Llama, 0.71 in GPT-5), enabling a Tier 3 autonomous-termination detector at 70% recall  $\times$  5% FP via combined  $v1 \cup v5$ . The signal validates across 3 model architectures from 3 labs but does NOT extend to short multi-task benchmarks (MALT null), scoping the claim. Mid-layer ablation reveals the cross-layer disagreement signal is edge-layer-driven, reframing WANDERING as “mid-layer-to-edge-layer alignment failure”. Three deployment tiers (forensics, advisory, autonomous) are all shippable.

## Reproducibility

All scripts and per-trajectory output JSONs are available at <https://github.com/OpenInterpretability/openinterp-swebench-harness> under Apache-2.0. Datasets used:

- OpenInterp Phase 6 (own)
- nebius/SWE-agent-trajectories (CC-BY-4.0)
- JetBrains-Research/agent-trajectories-swesmith-random-subset (HF public)
- SWE-bench/SWE-smith-trajectories (HF public)
- metr-evals/malt-public (HF gated, request approval)

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